

LABORATORY OBSERVATION/RECORD

Course:	Full Stack Web Development	Course Code:	23CM4121
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1. TITLE

Design a Static Webpage Using Pure Semantic HTML Components

2. AIM

To design a static webpage using only semantic HTML elements, without any CSS styling, and to demonstrate how browsers structure and render a document based purely on its markup.

3. OBJECTIVE

The objectives of this experiment are:

- To understand and apply semantic HTML5 elements such as **<header>**, **<nav>**, **<section>**, **<main>**, **<article>**, **<aside>** and **<footer>**.
 - To structure page layout using HTML tables for header and content columns.
 - To create a user inquiry form using native form controls (text, email, select, textarea, submit and reset).
 - To use lists (ordered and unordered) to present structured information.
 - To observe how a browser renders a page relying entirely on default styles.
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4. THEORY

Semantic HTML elements clearly describe their meaning to the browser and to the developer. Tags such as **<header>**, **<nav>**, **<section>**, **<main>**, **<article>**, **<aside>** and **<footer>** divide a page into meaningful regions instead of relying on generic **<div>** containers, improving accessibility and helping screen readers and search engines interpret page structure.

HTML tables (**<table>**, **<tr>**, **<td>**) were originally designed for tabular data but can also be used for basic page layout — arranging content into rows and columns — as demonstrated here for the header bar and the two-column article/sidebar layout.

HTML forms allow users to submit data using controls such as **<input type="text">**, **<input type="email">**, **<select>**, and **<textarea>**, wrapped inside a **<fieldset>** with a **<legend>**. The required attribute enables built-in browser-side validation without any JavaScript.

Because no CSS is used, the page relies entirely on the browser's default (user-agent) stylesheet for fonts, spacing and colors, which is why headings, links, and form controls appear in their standard unstyled form.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file and set up the basic document structure with **<!DOCTYPE html>**, **<html>**, **<head>** and **<body>**.
2. Inside **<header>**, create a table with the site title in one cell and a **<nav>** containing navigation links in another.
3. Add a horizontal rule (**<hr>**) to visually separate the header from the rest of the page.
4. Create a **<section>** for the hero/showcase area containing a heading, a short description and a link to the articles section.
5. Build the main two-column layout using a structural **<table>** with one row and two cells.

6. In the left cell (70% width), add a `<main>` element containing two `<article>` blocks describing structural tags and native form elements.
7. Inside the second article, add a `<form>` with a `<fieldset>` and `<legend>`, and include text, email, select and textarea inputs along with submit and reset buttons.
8. In the right cell (30% width, shaded background), add an `<aside>` with a heading, a paragraph, an unordered list and an ordered list.
9. Add a footer with a copyright notice centered using the `<center>` tag.
10. Open the HTML file in a browser and verify that all sections render correctly using only default browser styling.
11. Test the form fields by entering sample values and checking the built-in required-field validation.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Pure Semantic HTML Static Webpage</title>
</head>
<body>

<!-- Header Section -->
<header>
<table width="100%" border="0" cellspacing="0" cellpadding="10">
<tr>
<td>
<h1>StaticWeb</h1>
</td>
<td align="right">
<nav>
<a href="#home">Home</a> &nbsp;&nbsp;&nbsp;|&nbsp;&nbsp;&nbsp;
<a href="#articles">Articles</a> &nbsp;&nbsp;&nbsp;|&nbsp;&nbsp;&nbsp;
<a href="#about">About</a> &nbsp;&nbsp;&nbsp;|&nbsp;&nbsp;&nbsp;
<a href="#contact">Contact Us</a>
</nav>
</td>
</tr>
</table>
</header>
<hr>

<!-- Hero / Showcase Section -->
```

```

<section id="home">
<center>
<h2>Semantic Structures Without CSS</h2>
<p>This layout uses raw HTML components to build an organized
document hierarchy that browsers can cleanly parse out-of-the-box.</p>
<p><a href="#articles"><b>Read Our Articles Below &darr;</b></a></p>
</center>
</section>

<hr>

<!-- Main Content Layout using a Structural Table -->
<table width="100%" border="0" cellspacing="0" cellpadding="20">
<tr valign="top">
<!-- Left Side: Main Content Column -->
<td width="70%">
<main id="articles">
<article>
<h2>1. Core Structural Tags</h2>
<p>Published: July 7, 2026</p>
<p>When you omit cascading style sheets entirely,
the semantic markup handles all of the structural heavy
lifting. Elements like <code>&lt;main&gt;</code>,
<code>&lt;article&gt;</code>, and
<code>&lt;section&gt;</code> tell screen readers
and web scrapers exactly how information flows.</p>
<p>Using raw browser defaults ensures excellent loading
performance, perfect accessibility compliance, and
absolute structural clarity.</p>

</article>
<br><hr><br>
<article>
<h2>2. Native Interaction Elements</h2>
<p>Published: July 5, 2026</p>
<p>Browsers come equipped with excellent interactive
elements by default.</p>
<h3>Interactive Demonstration Forms</h3>
<form id="contact" action="#" method="post">
<fieldset>
<legend><b>User Inquiry Form</b></legend>
<p>
<label for="name">Full Name: </label>
<input type="text" id="name" name="username"
required placeholder="John Doe">
</p>
<p>
<label for="email">Email Address: </label>
<input type="email" id="email"
name="useremail" required>
</p>
<p>
<label for="topic">Inquiry Topic: </label>
<select id="topic" name="usertopic">
<option value="general">General Feedback</option>
<option value="technical">Technical Architecture</option>
<option value="academics">Academic Curriculum</option>
</select>

```

```

</p>
<p>
<label for="message">Message Details:</label><br>
<textarea id="message" name="usermessage"
rows="5" cols="40"
placeholder="Type your notes here..."></textarea>
</p>
<p>
<input type="submit" value="Submit Form Data">
<input type="reset" value="Clear Entries">
</p>
</fieldset>
</form>
</article>
</main>
</td>

<!-- Right Side: Sidebar Column -->
<td width="30%" bgcolor="#f4f4f4">
<aside id="about">
<h3>Document References</h3>
<p>Static pages map data out directly without
real-time database queries or processor overhead.</p>
<h4>Core Benefits:</h4>
<ul>
<li>Instant browser painting</li>
<li>Low memory footprints</li>
<li>High data compatibility</li>
</ul>
<h4>Required Components:</h4>

<ol>
<li>Document Type Definition</li>
<li>Semantic Wrapper Blocks</li>
<li>Data Input Control Interfaces</li>
</ol>
</aside>
</td>
</tr>
</table>
<hr>

<!-- Footer Section -->
<footer>
<center>
<p>&copy; 2026 Pure HTML Architecture.
Assembled completely without style configurations.</p>
</center>
</footer>
</body>
</html>

```

7. CODE EXPLANATION

Code	Explanation
<header>, <nav>, <footer>	Semantic elements marking the page's top banner, navigation links, and bottom footer region.

<section>, <main>, <article>, <aside>	Semantic containers dividing the page into a hero area, primary content, individual articles, and a sidebar.
<table>/<tr>/<td> (layout)	Used for positioning the header items and creating the two-column article/sidebar layout.
<fieldset> / <legend>	Groups related form controls under the "User Inquiry Form" with a visible caption.
required attribute	Triggers the browser's built-in validation — prevents form submission if Name or Email fields are empty.
<select> / <option>	Provides a dropdown of predefined inquiry topics for the user to choose from.
<textarea>	Provides a multi-line text input for the message details.
<input type="submit"> <input type="reset">	Native buttons to submit the form data or clear all entered values.
/	Unordered and ordered lists used in the sidebar to present benefits and required components.

8. TEST CASES AND EXECUTION DATA

Each section and form control was verified by loading the page in a browser and interacting with it.

Test Case	Feature Tested	Expected Result	Actual Result	Status
TC-01	Page load	Header, hero, articles, sidebar and footer rendered correctly	Header, hero, articles, sidebar and footer rendered correctly	Pass
TC-02	Navigation links	Home, Articles, About and Contact Us links visible and functional	Home, Articles, About and Contact Us links visible and functional	Pass
TC-03	Two-column layout	Articles 70% width; sidebar 30% with shaded background as expected	Articles 70% width; sidebar 30% with shaded background as expected	Pass
TC-04	Form – required fields	Submitting with Name or Email empty blocked by browser	Submitting with Name or Email empty blocked by browser	Pass
TC-05	Form – dropdown & text area	Inquiry Topic selectable; Message Details prompt and field text accepted	Inquiry Topic selectable; Message Details prompt and field text accepted	Pass
TC-06	Reset button	Clicking Clear Entries resets all form fields	Clicking Clear Entries resets all form fields	Pass

9. OUTPUT

Output: Webpage Rendered in Browser

The rendered page shows the two-column layout with articles on the left and the sidebar on the right. The User Inquiry Form is visible with all fields — Full Name, Email Address, Inquiry Topic and Message Details — along with Submit and Clear Entries buttons, all using unstyled default browser formatting.

1. Core Structural Tags

Published: July 7, 2026

When you omit cascading style sheets entirely, the semantic markup handles all of the structural heavy lifting. Elements like `<main>`, `<article>`, and `<section>` tell screen readers and web scrapers exactly how information flows. Using raw browser defaults ensures excellent loading performance, perfect accessibility compliance, and absolute structural clarity.

2. Native Interaction Elements

Published: July 5, 2026

Browsers come equipped with excellent interactive elements by default. We can capture text inputs, provide multiple-choice configurations, and submit structured information to backend parsers using pure markup.

Interactive Demonstration Forms

User Inquiry Form

Full Name:

Email Address:

Inquiry Topic:

General Feedback

Message Details:

ibbbbu

Submit Form Data

Clear Entries

10. RESULT

Thus, a static webpage was successfully designed using pure semantic HTML components, including **header**, **navigation**, **section**, **main**, **article**, **aside** and **footer** elements, a table-based layout, and a native HTML form with required-field validation. The page rendered correctly in the browser using only default styling, and all sections and form controls were verified as working as expected.
